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CERTIFICATE

This is to certify that Ms. /Mr. ***Roy Arpit ,Tanya baid,Shivani ram*** has satisfactorily completed activity-1 prescribed by JAIN (Deemed to be University) for the 5th Semester degree course in the year 2026-2027 **Presentation** on the topic **COLLEGE EQUIPMENT / ASSET TRACKING SYSTEM – CampusTrack** for Activity – 1.

Signature of the staff member
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1. Business / Startup Name

CampusTrack – Intelligent College Equipment & Asset Tracking System

Tagline: "Every Asset, Always Accounted For"

CampusTrack is a QR-code based digital asset and maintenance management system designed to help colleges, schools and training institutes track, allocate and maintain their equipment — from projectors and laptops to lab instruments and networking hardware — in a simple, transparent and centralized manner.

2. Product / Service Name

CampusTrack – QR-Based College Equipment & Asset Tracking System

CampusTrack is a web and mobile-based platform that assigns a unique QR code to every institutional asset. Scanning the QR code instantly reveals the asset's identity, current location, assignment, maintenance history and condition, and allows staff to raise a maintenance request in a single tap.

The system uses a centralized database and a rule-based/AI-assisted maintenance engine to schedule preventive maintenance, generate tickets for reported faults, and route them to the correct maintenance team automatically.

3. Problem Statement – What Problem Are We Solving?

Colleges, schools and training institutes own hundreds to thousands of physical assets, including:

- Projectors and smart boards
- Laptops and desktop computers
- Laboratory equipment and instruments
- Cameras and audio-visual equipment
- Microphones and public-address systems
- Routers, switches and networking hardware
- Furniture and other departmental assets

A major challenge is that most institutions still track these assets through manual registers or scattered spreadsheets. This makes it difficult to know:

- Who currently has custody of an asset
- Which room, lab or department an asset is assigned to
- When an asset was last serviced and when it is next due for maintenance
- Whether a reported fault has actually been fixed
- How many assets are idle, damaged or missing during an audit

For example, a projector reported as "not working" in one department may take days to reach the right maintenance staff simply because there is no structured way to log, route and track the complaint.

Problem We Address

The core problem is the lack of a centralized, QR-based asset register and maintenance-ticketing system that gives institutions real-time visibility into what they own, where it is, its condition, and its maintenance status.

4. Our Solution – How Will We Solve the Problem?

CampusTrack assigns every physical asset a unique, tamper-resistant QR code sticker at the time of onboarding. Scanning the code with a phone or the CampusTrack app instantly displays the asset's full profile and lets staff take action.

Example

A lab assistant scans the QR code on a projector:

- Asset ID: P-104
- Currently assigned to: Lab 3
- Last maintenance: 12 Aug
- Next maintenance due: 12 Nov
- Condition: Good

If the projector develops a fault, the same staff member taps “Report Issue” and types: “Projector isn't working.” CampusTrack automatically:

- Creates a maintenance ticket linked to Asset P-104
- Tags the location (Lab 3) and category (AV Equipment)
- Assigns it to the correct maintenance/IT team
- Notifies the admin and the reporting staff member
- Tracks the ticket until it is marked resolved and verified

This removes the need for phone calls, physical complaint books or informal follow-ups, and gives the institution a permanent, searchable history for every asset it owns.

5. Product / Service Description

CampusTrack consists of two major interfaces:

A. Staff / Faculty Mobile App

- Register / login with institutional ID
- Scan an asset's QR code to view its full profile
- Report a fault or issue with photo attachment
- Track the status of a reported issue
- View assets assigned to their department or room
- Receive notifications on ticket updates and maintenance schedules

B. Admin / Facilities Dashboard

- Add, edit and generate QR codes for new assets
- View the complete asset inventory, searchable and filterable

- Assign or reassign assets to departments, rooms or staff
- View and manage all maintenance tickets
- Schedule preventive maintenance and receive due-date alerts
- Track technician workload and response time
- Generate audit-ready reports (for NAAC/NBA or internal audits)
- View asset distribution and condition analytics

Asset / Ticket Status Flow

Reported → Ticket Created → Assigned to Technician → In Progress → Resolved → Verified by Staff

This gives both staff and administrators complete visibility into the life cycle of every asset and every complaint.

6. Target Customers

CampusTrack has two main user groups.

Primary Users

Faculty, Lab Assistants and Administrative Staff

They use CampusTrack daily to look up asset details and report faults in the labs, classrooms and offices they work in.

Institutional Customers

Colleges, Universities and Training Institutes

Institutions purchase CampusTrack to maintain a centralized, auditable record of every physical asset they own and to streamline maintenance operations.

Other Potential Users

- Schools and coaching institutes
- Corporate training centres
- Co-working spaces and libraries
- Hostels and campus facility-management teams

7. Unique Selling Proposition (USP)

The USP of CampusTrack is:

"Scan an asset, know everything about it — and fix it faster."

Traditional asset registers are static spreadsheets that quickly go out of date. CampusTrack adds a live, scannable layer on top of every physical asset.

What Makes CampusTrack Different?

- Instant identification: One QR scan reveals full asset history — no manual lookup needed
- One-tap fault reporting: Staff report issues directly from the asset itself
- Automated maintenance scheduling: Preventive maintenance alerts based on due dates
- Centralized accountability: Every asset has a clear, current custodian and location
- Audit-ready reporting: Instantly exportable records for accreditation and internal audits
- Analytics for administrators: Identify high-fault equipment and departments with repeated issues

8. Market Need & Opportunity

Educational institutions are under increasing pressure to demonstrate proper infrastructure management, both for accreditation bodies (such as NAAC/NBA) and for internal budget accountability. Most currently rely on manual registers that cannot scale with the number of assets they own.

Potential Market

- Colleges and universities
- Schools and coaching institutes
- Polytechnic and vocational training institutes
- Corporate training academies

Expansion Opportunity

Once proven in one institution, CampusTrack can be customized for different types of campuses:

- College A → Labs + AV Equipment + Computers
- College B → Furniture + Networking + Library Equipment
- Training Institute → Studio/Recording + Computers + Instruments

Therefore, the same core asset-and-ticketing engine can support many different institutional deployments.

9. Business Model – How Will We Make Money?

CampusTrack will follow a B2B SaaS model, where educational institutions pay to use the platform.

Revenue Sources

1. Institutional Subscription

Institutions pay a monthly or yearly subscription based on the number of assets and users on the platform.

2. Setup & Onboarding Fee

- QR code generation and printing for existing assets
- Bulk asset data upload and categorization
- Dashboard configuration for departments and roles
- Staff training and onboarding

3. Premium Analytics

- Maintenance trend and downtime analysis

- Department-wise fault frequency reports
- Asset utilization and lifecycle analytics
- Predictive maintenance recommendations

4. Maintenance & Support Packages

Annual technical support and system-maintenance packages provide recurring revenue.

Example Revenue Model

Suppose one institution pays ₹15,000 per month:

Annual revenue = ₹15,000 × 12 = ₹1,80,000

If the platform serves 20 institutions:

₹1,80,000 × 20 = ₹36,00,000 per year

These are illustrative business projections rather than guaranteed revenue.

10. Marketing & Promotion Strategy

CampusTrack will primarily use institutional marketing, since colleges, schools and training institutes are the main customers.

Marketing Methods

1. Demonstration to College Administration

Conduct live demos showing how quickly an asset can be scanned, identified and a fault reported.

2. Pilot Programs

Offer a free or discounted pilot in one department or lab block to prove value before full rollout.

3. Website & Social Media

Maintain a professional website and LinkedIn presence targeted at facilities managers and college administrators.

4. College and Technology Events

Showcase CampusTrack at hackathons, innovation fairs and education-technology exhibitions.

5. Partnerships

Partner with education-technology resellers, campus-management software vendors and facility-management companies.

6. Referral and Institutional Networking

Use references from satisfied institutions (e.g., affiliated or sister colleges) to reach new customers.

11. Technology and Innovation Used

CampusTrack uses a combination of QR/barcode technology, web and mobile development, database management and data analytics to create an intelligent asset-tracking system.

Frontend Technology

The admin dashboard is built using React.js, and the staff-facing mobile app uses React Native, allowing QR scanning directly through a phone camera.

Backend Technology

The backend is developed using Node.js and Express.js, handling asset records, ticket creation, department routing and notifications.

Database

PostgreSQL stores asset details, QR-code mappings, maintenance history, ticket records, user roles and departmental assignments.

QR Code Technology

Each asset receives a unique QR code generated using standard QR libraries and printed on a durable, weather-resistant sticker. Scanning the code retrieves the asset's live record from the database.

Example

Scan Result for Asset P-104:

- Category → AV Equipment
- Assigned Location → Lab 3
- Condition → Good
- Next Maintenance → 12 Nov

Maintenance Intelligence

Instead of treating every reported fault the same way, the system can consider factors such as asset criticality, number of similar complaints, and how long the asset has been due for service to prioritise which tickets need urgent attention.

Future AI Innovation

The system can later be enhanced with image recognition to auto-detect visible damage from uploaded photographs, predictive maintenance using historical failure data, and IoT/RFID sensors for real-time location tracking of movable assets.

Main Innovation

The key innovation of CampusTrack is that it does not just store a list of assets — it converts every physical item into a scannable, trackable and self-reporting record, turning informal complaint books into structured, actionable maintenance data.

12. Estimated Cost

Developing the CampusTrack prototype requires spending on software development, hosting, database services, QR code generation and printing, mobile app development and testing. The following is an estimated initial prototype budget.

Expense	Estimated Cost
Domain and Website	₹2,000
Cloud / Server Hosting	₹6,000
Database and Storage	₹3,000
QR Code Generation & Printing	₹2,500
Mobile App Development Tools	₹4,500
Testing and Miscellaneous	₹3,000
Marketing and Demonstration	₹4,000
Total Estimated Cost	₹25,000

Explanation of Costs

Domain and Website – ₹2,000: Registering a domain and setting up CampusTrack's basic web presence.

Cloud / Server Hosting – ₹6,000: Running the web dashboard and backend services so institutions can access the platform online.

Database and Storage – ₹3,000: Maintaining asset records, QR mappings, maintenance history and uploaded images.

QR Code Generation & Printing – ₹2,500: Generating and printing durable QR stickers for pilot assets.

Mobile App Development Tools – ₹4,500: Development and testing tools for the staff scanning app.

Testing and Miscellaneous – ₹3,000: Debugging, QA and small technical expenses.

Marketing and Demonstration – ₹4,000: Creating demo material, presentations and pilot testing with a partner institution.

Total Initial Investment

The estimated initial investment for developing and demonstrating the CampusTrack prototype is approximately ₹25,000. This is an illustrative prototype estimate, not a fixed commercial development cost.

13. Selling Price & Expected Profit

CampusTrack will use a subscription-based pricing model.

Example Pricing

- Basic Plan: ₹10,000/month – for small institutions with a limited number of assets
- Professional Plan: ₹20,000/month – includes maintenance scheduling, analytics and multi-department dashboards
- Enterprise Plan: Customized pricing – for large campuses requiring integrations with existing ERP systems

Example Calculation

Suppose one institution chooses the Professional Plan:

Monthly revenue = ₹20,000

If the estimated monthly operating cost is ₹12,000:

Estimated monthly operating surplus = ₹20,000 − ₹12,000 = ₹8,000

Estimated annual operating surplus = ₹8,000 × 12 = ₹96,000

These figures are illustrative estimates and would change based on actual infrastructure, staffing and operational costs.

14. Expected Impact

Impact on Institutions

- Reduced equipment loss and misplacement
- Clear accountability for every asset
- Faster fault resolution and reduced classroom/lab downtime
- Audit-ready records for accreditation and internal reviews
- Data-driven decisions on equipment replacement and budgeting

Impact on Staff and Students

- Functioning labs and classrooms with fewer disruptions
- Simple, one-tap way to report a problem instead of informal complaints
- Transparency on when a reported issue will be fixed

Institutional Impact

By converting scattered asset information into structured data, institutions can identify recurring problem areas — such as a lab with frequent projector failures — and plan maintenance or replacement proactively.

15. Future Scope

- IoT/RFID Tracking: Real-time location tracking of movable assets across campus
- Predictive Maintenance: Using historical fault data to predict which assets are likely to fail next
- Image-Based Damage Detection: AI analysis of uploaded photos to auto-assess damage severity
- Mobile Push Notifications: Automatic reminders for upcoming maintenance and ticket updates
- ERP Integration: Connecting CampusTrack with existing college administration or library systems
- Multi-Campus Support: Managing assets across multiple campuses of the same institution from one dashboard
- Expansion to Corporate & Healthcare: Adapting the same engine for offices and hospital equipment tracking

16. Why Should Someone Invest in Our Idea?

CampusTrack combines QR technology, asset management and maintenance automation into one practical platform built specifically for the way colleges actually operate.

The system addresses a real, recurring operational problem: keeping track of hundreds of physical assets and resolving faults quickly. Instead of a static register, CampusTrack adds a live, scannable and self-reporting layer to every asset.

Its software-based, subscription model allows the platform to be customized for different institutions without rebuilding the system from scratch, giving it strong potential to scale across colleges, schools and training institutes.

Final Pitch

CampusTrack turns every projector, laptop and lab instrument into a smart, trackable asset. By connecting staff, administrators and maintenance teams through one QR-based platform, CampusTrack aims to make campus infrastructure management more organized, accountable and efficient.